# CP1.1: Complex Numbers

## 1. Introduction and Definition

**Imaginary unit:** The imaginary unit $i$ is defined by $i^2 = -1$, so $i = \sqrt{-1}$.

**Complex number:** A complex number has the form $z = a + bi$ where $a, b \in \mathbb{R}$. The set of all complex numbers is denoted $\mathbb{C}$.

**Real part:** $\mathrm{Re}(z) = a$
**Imaginary part:** $\mathrm{Im}(z) = b$ (note: this is the real number $b$, not $bi$)

Every real number is also a complex number with imaginary part zero. Numbers of the form $bi$ (where $b \neq 0$) are called *purely imaginary*.

### Powers of i

The powers of $i$ cycle with period 4:

| Power | $i^1$ | $i^2$ | $i^3$ | $i^4$ | $i^5$ |
|---|---|---|---|---|---|
| Value | $i$ | $-1$ | $-i$ | $1$ | $i$ |

To evaluate $i^n$: find the remainder when $n$ is divided by 4 and use the table above.

**Exam tip:** When you see $\sqrt{-k}$ for $k > 0$, write it as $i\sqrt{k}$. Never leave a negative under a square root in your final answer. For example: $\sqrt{-9} = 3i$, not $3\sqrt{-1}$.

## 2. Arithmetic with Complex Numbers

## Addition and Subtraction

Add or subtract real and imaginary parts separately:

$$(a + bi) \pm (c + di) = (a \pm c) + (b \pm d)i$$

## Multiplication

Expand using FOIL (like with surds), then substitute $i^2 = -1$:

$$(a + bi)(c + di) = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i$$

**Example:** Find $(3 + 2i)(1 - 4i)$.

$$(3 + 2i)(1 - 4i) = 3(1) + 3(-4i) + 2i(1) + 2i(-4i)$$
$$= 3 - 12i + 2i - 8i^2$$
$$= 3 - 10i - 8(-1)$$
$$= 3 - 10i + 8 = 11 - 10i$$

## The Complex Conjugate

**Complex conjugate:** If $z = a + bi$, then the conjugate is $z^* = a - bi$ (also written $\bar{z}$).

Key properties:

- $z + z^* = 2a$ (always real)
- $z \cdot z^* = a^2 + b^2$ (always real and non-negative)
- $(z^*)^* = z$
- $(z_1 z_2)^* = z_1^* z_2^*$
- $(z_1 + z_2)^* = z_1^* + z_2^*$

## Division

To divide complex numbers, multiply both numerator and denominator by the conjugate of the denominator:

$$\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{(c+di)(c-di)} = \frac{(ac+bd)+(bc-ad)i}{c^2+d^2}$$

The denominator $c^2 + d^2$ is always real, giving a result in the form $x + yi$.

**Example:** Simplify $\dfrac{3+5i}{1+2i}$.

Multiply top and bottom by conjugate $(1-2i)$:

$$\frac{(3+5i)(1-2i)}{(1+2i)(1-2i)} = \frac{3-6i+5i-10i^2}{1-4i^2} = \frac{3-i+10}{1+4} = \frac{13-i}{5} = \frac{13}{5} - \frac{1}{5}i$$

**Exam tip:** Always check your division: multiply your answer by the original denominator and verify you get the numerator back. Losing marks on arithmetic in division is very common.

# 3. Modulus and Argument

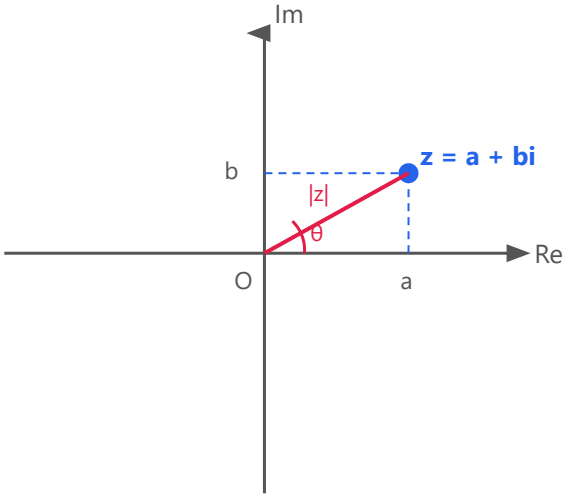
**Modulus:** $|z| = \sqrt{a^2+b^2}$ — this is the distance of $z$ from the origin in the Argand diagram.

**Argument:** $\arg(z) = \theta$ where $\tan\theta = \dfrac{b}{a}$ — this is the angle measured anticlockwise from the positive real axis.

The **principal argument** satisfies $-\pi < \arg(z) \leq \pi$.

## Argand Diagram

A complex number $z = a + bi$ is represented as a point $(a, b)$ in the complex plane, where the horizontal axis is the real axis and the vertical axis is the imaginary axis.




## Calculating the Argument Carefully

Use $\tan\theta = b/a$ but you must place the angle in the correct quadrant. Always draw a sketch first:

| Quadrant | Sign of $a$, $b$ | Principal argument range | Method |
|---|---|---|---|
| 1st | $a > 0, b > 0$ | $0 < \theta \leq \frac{\pi}{2}$ | $\theta = \arctan\left(\frac{b}{a}\right)$ |
| 2nd | $a < 0, b > 0$ | $\frac{\pi}{2} < \theta \leq \pi$ | $\theta = \pi - \arctan\left(\frac{b}{\lvert a\rvert}\right)$ |
| 3rd | $a < 0, b < 0$ | $-\pi < \theta \leq -\frac{\pi}{2}$ | $\theta = -\pi + \arctan\left(\frac{\lvert b\rvert}{\lvert a\rvert}\right)$ |
| 4th | $a > 0, b < 0$ | $-\frac{\pi}{2} < \theta < 0$ | $\theta = -\arctan\left(\frac{\lvert b\rvert}{a}\right)$ |

**Example:** Find $|z|$ and $\arg(z)$ for $z = -1 + \sqrt{3}\,i$.

$$|z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$$

$z$ is in the second quadrant ($a < 0, b > 0$).
Reference angle: $\arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$
Therefore $\arg(z) = \pi - \frac{\pi}{3} = \dfrac{2\pi}{3}$

**Exam tip:** The most common error is computing $\arctan(b/a)$ and forgetting to adjust for quadrant. Always establish the quadrant FIRST, then compute the reference angle, then adjust.

### Modulus-Argument Properties

$$|z_1 z_2| = |z_1||z_2| \qquad \arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

$$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|} \qquad \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$

(Arguments are taken modulo $2\pi$ to keep them in the principal range $(-\pi, \pi]$.)

## 4. Solving Quadratic Equations with Complex Roots

When a quadratic $az^2 + bz + c = 0$ has a negative discriminant ($b^2 - 4ac < 0$), the roots are complex conjugates of each other.

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

**Worked Example:** Solve $z^2 + 4z + 13 = 0$.

Using the quadratic formula with $a = 1, b = 4, c = 13$:

$$z = \frac{-4 \pm \sqrt{16 - 52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2} = -2 \pm 3i$$

So the roots are $z = -2 + 3i$ and $z = -2 - 3i$.

**Verification:** Sum of roots $= (-2+3i)+(-2-3i) = -4 = -b/a$ ✓
Product of roots $= (-2+3i)(-2-3i) = 4+9 = 13 = c/a$ ✓

**Conjugate Root Theorem:** If the coefficients of a polynomial are all real, then complex roots occur in conjugate pairs. That is, if $z = p+qi$ is a root, then $z^* = p-qi$ is also a root.

# 5. Solving Cubic and Quartic Equations

## Cubics with Complex Roots

A cubic with real coefficients has either:

- Three real roots, or
- One real root and one conjugate pair of complex roots

**Strategy:** If given one complex root $\alpha$, immediately write down $\alpha^*$ as another root. The factor $(z-\alpha)(z-\alpha^*)$ is a real quadratic. Divide the cubic by this quadratic (polynomial long division or compare coefficients) to find the third root.

**Example:** Given that $2+i$ is a root of $z^3-7z^2+17z-15=0$, find all roots.

Since coefficients are real, $2-i$ is also a root.

Form the quadratic factor:

$$(z-(2+i))(z-(2-i)) = (z-2)^2+1 = z^2-4z+5$$

Divide: $z^3-7z^2+17z-15 \div (z^2-4z+5)$

$z^3-7z^2+17z-15 = (z^2-4z+5)(z-3)$

So the three roots are $z=2+i,\ z=2-i,\ z=3$.

## Quartics with Complex Roots

A quartic with real coefficients has either:

- Four real roots
- Two real roots and one conjugate pair
- Two conjugate pairs of complex roots

**Strategy:** If given one complex root $\alpha$, use $\alpha^*$ to form one quadratic factor. Divide the quartic by it to get another quadratic, then solve that for the remaining roots.

**Example:** Given that $1+2i$ is a root of $z^4 - 2z^3 + 6z^2 - 2z + 5 = 0$, find all roots.

Conjugate pair: $1+2i$ and $1-2i$.
Quadratic factor: $(z-1-2i)(z-1+2i) = (z-1)^2 + 4 = z^2 - 2z + 5$

Divide: $z^4 - 2z^3 + 6z^2 - 2z + 5 = (z^2 - 2z + 5)(z^2 + 1)$

Solve $z^2 + 1 = 0 \Rightarrow z^2 = -1 \Rightarrow z = \pm i$

All four roots: $1+2i,\ 1-2i,\ i,\ -i$

### Finding a Root by Substitution

If asked to verify or find a complex root directly, substitute into the polynomial and set real and imaginary parts equal to zero. This gives a system of equations.

**Example:** Find $a$ and $b$ if $z = 1+2i$ is a root of $z^2 + az + b = 0$ (where $a, b \in \mathbb{R}$).

Sum of roots: $(1+2i) + (1-2i) = 2 = -a \Rightarrow a = -2$
Product of roots: $(1+2i)(1-2i) = 5 = b \Rightarrow b = 5$

## 6. Equating Real and Imaginary Parts

Two complex numbers are equal if and only if their real parts are equal AND their imaginary parts are equal:

$$a + bi = c + di \iff a = c \text{ and } b = d$$

**Example:** Solve $z^2 = 5 + 12i$.

Let $z = x + yi$. Then $(x + yi)^2 = x^2 - y^2 + 2xyi = 5 + 12i$.

Equating real parts: $x^2 - y^2 = 5$ ... (1)
Equating imaginary parts: $2xy = 12$, so $y = \frac{6}{x}$ ... (2)

Substituting (2) into (1): $x^2 - \frac{36}{x^2} = 5$
$x^4 - 5x^2 - 36 = 0 \Rightarrow (x^2 - 9)(x^2 + 4) = 0$
Since $x \in \mathbb{R}$: $x^2 = 9 \Rightarrow x = \pm 3$

When $x = 3$: $y = 2 \to z = 3 + 2i$
When $x = -3$: $y = -2 \to z = -3 - 2i$

**Exam tip:** When finding square roots of complex numbers by letting $z = x + yi$ and equating parts, remember that you always get two answers (a conjugate pair). Both must be stated.

## 7. Key Results Summary

| Property | Formula |
|---|---|
| Definition of $i$ | $i^2 = -1$ |
| Conjugate | $z^* = a - bi$ if $z = a + bi$ |
| Modulus | $\lvert z \rvert = \sqrt{a^2 + b^2}$ |
| Product × Conjugate | $zz^* = \lvert z \rvert^2 = a^2 + b^2$ |
| Argument (principal) | $-\pi < \arg(z) \leq \pi$ |

| | |
|---|---|
| Modulus of product | $\lvert z_1 z_2 \rvert = \lvert z_1 \rvert \lvert z_2 \rvert$ |
| Argument of product | $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ |
| Conjugate root theorem | Real-coefficient polynomials: complex roots in conjugate pairs |

## 8. Practice Checklist

- Can you add, subtract, multiply, and divide complex numbers in $a + bi$ form?
- Can you find the modulus and principal argument of any complex number?
- Can you solve quadratics with negative discriminant?
- Given one complex root of a cubic/quartic, can you find all other roots?
- Can you find square roots of complex numbers using the "let $z = x + yi$" method?
- Can you use the conjugate root theorem to form factor pairs?

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